

Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects

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The problem

The accuracy–interpretability tradeoff is usually treated as fundamental: either fit a linear model you can read, or fit a black box you cannot. We argue it is an artifact of *how* black boxes are read. A logistic regression is interpretable not because it is linear but because it reports a table of effects with standard errors. Any model can be made to do that.

The estimand

For a fitted prediction function $\hat{f}$ and feature x_j , the average marginal effect is the average partial derivative over the covariate distribution,

$$\text{AME}_j = \mathbb{E} \left[\frac{\partial \hat{f}(\mathbf{x})}{\partial x_j} \right].$$

This is standard in the econometrics of nonlinear models. It is well defined for *any* supervised learner, and reduces to the usual coefficient when $\hat{f}$ is a generalized linear model.

The estimator

Analytical derivatives are replaced by centered finite differences with an adaptive step size,

$$\widehat{\text{AME}}_j = \frac{1}{n} \sum_{i=1}^n \frac{\hat{f}(\mathbf{x}_i + h_j \mathbf{e}_j) - \hat{f}(\mathbf{x}_i - h_j \mathbf{e}_j)}{2h_j},$$

with the step size scaled to each feature and floored on the integers,

$$h_j = \begin{cases} \max(0.05 \hat{\sigma}_j, \frac{1}{2}) & x_j \text{ integer-valued,} \\ 0.05 \hat{\sigma}_j & \text{otherwise.} \end{cases}$$

- The integer floor matters for tree ensembles: predictions are piecewise constant, and a step that never crosses a split threshold returns a derivative of exactly zero.
- Binary features use the contrast $\hat{f}(\mathbf{x}^{(j,1)}) - \hat{f}(\mathbf{x}^{(j,0)})$ instead.

Inference

Standard errors come from a nonparametric bootstrap over $(\mathbf{x}_i, y_i)$ pairs. Each replicate is **refit from a fresh initialization** on the resample and evaluated there.

- Refitting from scratch, rather than warm-starting from $\hat{f}$, is what makes the replicate spread reflect how far the fitted function actually moves with the data.
- Output is a coefficient table — estimate, standard error, significance — with no restriction on the form of $\hat{f}$.

Effects agree across model classes

UCI Adult Income ($n = 48,842$), full specification. AME on $P(\text{income} > \$50\text{k})$.

	Logistic	Forest	XGBoost	Net
education	0.031	0.029	0.031	0.028
married	0.275	0.245	0.236	0.239
female	−0.027	−0.022	−0.012	−0.027
McFadden R^2	0.400	0.450	0.492	0.455

Four learners that differ substantially in fit report the same effect of education to within **0.3 percentage points**. This comparison is only possible because the AME puts every model on a common scale.

Effects in the units of the problem

Ames Housing ($n = 1,460$). AME on sale price, in dollars.

	Linear	Forest
Living area (/sq ft)	\$66 (13.06)	\$56 (5.25)
Overall quality (/pt)	\$16,465 (1,287)	\$15,636 (1,643)

Standard errors in parentheses. The ensemble is not uniformly more precise: it wins where the linear model must fit a nonlinear relationship with one slope, and loses where the relationship is nearly linear.

Why not SHAP?

SHAP values, PDP slopes and LIME coefficients are *descriptive*. There is no standard error, no confidence interval, no hypothesis test.

They are also not on the scale of the outcome. Adult, logistic, full specification:

	AME	SHAP	PDP
age	0.0027	− 0.0270	0.0028
education	0.0306	− 0.1186	0.0269
married	0.2752	−0.0768	0.2749
female	−0.0269	0.0022	−0.0270

The AME and the PDP slope agree to three decimals on three of four features — as they should, since both are averaging the same surface. SHAP disagrees in **magnitude and in sign**: it reports age and education as pushing income *down* in a model whose coefficients on both are positive and precisely estimated. For above-ground living area in Ames, the SHAP attribution is roughly **50×** the AME for the same linear model — not interpretable as a marginal effect in any standard sense.

Simulation 1: the estimator recovers the truth

$x_1, \dots, x_4 \sim N(0, 1)$ independently; $\eta = 2x_1 + 3x_2$; x_3, x_4 are noise with a true effect of zero. 500–1,000 Monte Carlo iterations.

- Bias is negligible for every learner at every sample size on the noise features.
- Logistic and the network recover the signal effects at the $O(n^{-1/2})$ rate.
- Random forests develop a **negative bias that does not vanish** with n — regularization from depth capping and subsampling.

Simulation 2: where the bootstrap fails

Coverage of nominal 95% intervals, linear classification design.

	$n=250$	$n=1,000$	$n=5,000$
Logistic	0.934	0.930	0.924
XGBoost	0.938	0.888	0.994
Forest	0.992	0.978	0.802

The failure is bias, not variance. The forest’s bootstrap standard error is *conservative* — 1.2 to 1.5× the true sampling deviation — so the intervals are, if anything, too wide.

What breaks is that the bias stays put while the interval shrinks:

Forest, x_1	$n=250$	$n=1,000$	$n=5,000$
bias	0.0126	−0.0018	−0.0157
std. error	0.0608	0.0227	0.0074
bias / s.e.	0.21	0.08	2.11

At $n = 5,000$ the half-width of the interval (0.0143) is smaller than the bias (0.0157). XGBoost, whose bias *does* vanish with n , never fails in that direction — so the failure is specific to learners whose regularization bias does not shrink.

Takeaways

- Any supervised model can report a coefficient table on the scale of the outcome, with inference.
- Effects are stable across learners that differ substantially in fit.
- The refitting bootstrap is valid where the learner’s regularization bias is negligible, and **fails silently where it is not**.
- Open: bootstrap validity is conjectured, not proved. A companion journal paper establishes gradient consistency for neural networks, gradient boosting and Mondrian forests, and replaces the bootstrap with a debiased, cross-fitted, Neyman-orthogonal estimator that restores nominal coverage for Breiman forests.

Software

`marginfx` — `pip install marginfx`
One call per model. scikit-learn, XGBoost, TensorFlow and PyTorch.